

# TrES-1b

R-0271

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-1\_170612 · created 2026-07-17T13:01:49+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457916.7446 +/- 0.0017 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.138 +/- 0.016                 |
| Transit depth (Rp/Rs)^2                           | 1.91 +/- 0.45 [%]               |
| Orbital Inclination (inc)                         | 88.34 +/- 1.19                  |
| Airmass coefficient 1 (a1)                        | 1.204 +/- 0.02                  |
| Airmass coefficient 2 (a2)                        | -0.1321 +/- 0.005               |
| Scatter in the residuals of the lightcurve fit is | 1.67 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 3.41                            |
| Optimal Annulus                                   | 5.21                            |
| Transit Duration (day)                            | 0.0987 +/- 0.0073               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.138 ± 0.016 vs 0.1358 (deviation 0.11σ) |
| Duration fit vs published | 0.0987 d vs 0.1042 d (deviation 0.59σ)    |
| Mid-time O–C              | -2.5 min                                  |
| Depth SNR                 | 6.8                                       |
| Residual scatter          | 1.67 %                                    |

## Light curve

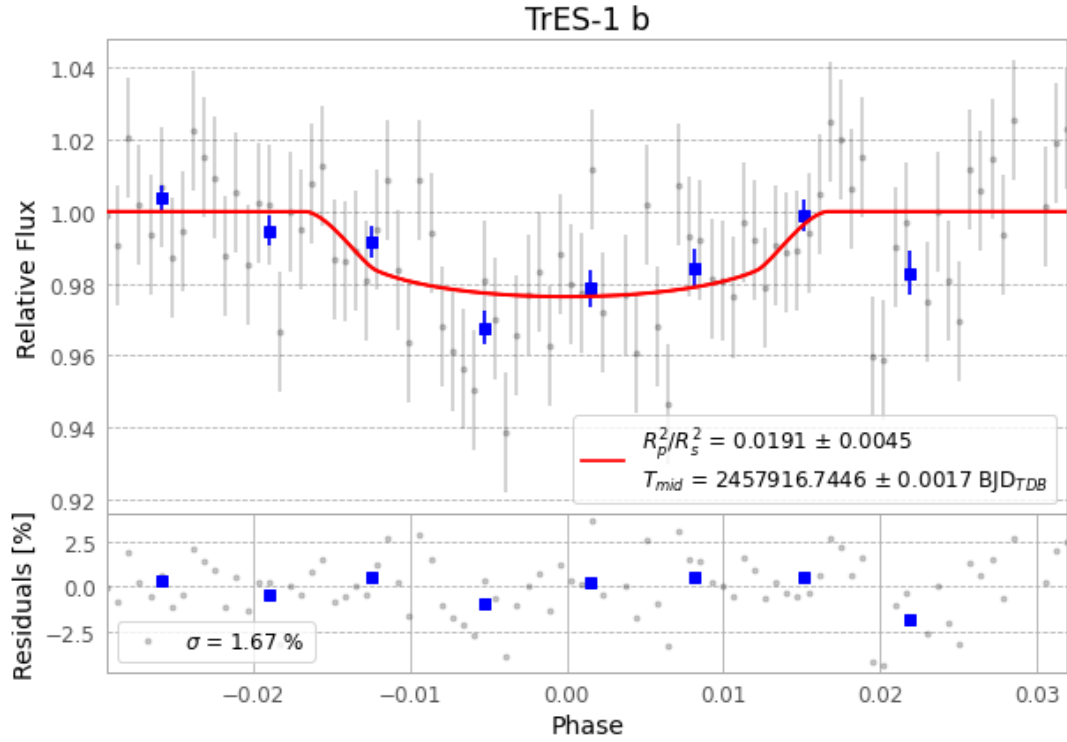


Figure 1 — FinalLightCurve\_TrES-1 b\_2017-06-11.png (SHA-256 9ad8ae608546df10...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [234, 198], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 ef0af9d7111ea9d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1cb2214

## Data provenance

90 FITS frames (59 MB), TRES-1170612033912.FITS → TRES-1170612080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-170612.log (7502c36d300b...), FinalLightCurve\_TrES-1 b\_2017-06-11.png (9ad8ae608546...), FinalLightCurve\_TrES-1 b\_2017-06-11.csv (89e3bc32251d...)

## Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 13:01Z.